

Validated-by-Design

*A Shift-Left Approach to Computer System Validation for Cloud-Native
GxP Platforms*

Position Paper · Paper 1 of 3 — Vision & Operating Model

Authors: Corrado Iorizzo, Steven Schmidt, Gupta Telluri

Audience: Executive sponsors · Business Quality and GxP leaders · Business Operational Excellence
leaders · IT Quality and GxP leaders · IT Data Platform leaders

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About This Paper

This is Paper 1 of a three-part series on Shift-Left Computer System Validation, written so each paper can be read on its own.

The shared thesis across the series: retrospective CSV practices can struggle to keep assurance aligned with cloud-native GxP platforms; the proposed response is to engineer governed controls and evidence capture into the architecture, while retaining intended-use assessment, risk-based judgement, exceptions, change impact, and Quality accountability.

The series is organised by audience:

- Paper 1 — Vision & Operating Model (this paper): for executive sponsors and Business and IT leaders across Quality, GxP, Operational Excellence, and Data Platforms. It explains the vision, operating model, governance, adoption roadmap, and grounded conclusion.
- Paper 2 — Technical Reference Architecture: for architects and platform teams. It develops the architecture principles, policy-as-code, lineage and consistency patterns, and reference architecture.
- Paper 3 — Regulatory & Implementation Guide: for CSV, Quality, and Regulatory teams. It develops regulatory mapping, implementation and inspection readiness, AI workloads, and risks and mitigations.

This paper stays deliberately concise so it can be read end-to-end and shared. Readers who need the technical or regulatory depth are pointed to Papers 2 and 3.

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1. Executive Summary

Retrospective CSV practices can struggle to keep assurance aligned with cloud-native GxP platforms; Shift-Left CSV proposes engineering governed controls and evidence capture into the platform while preserving workload-specific assessment and Quality accountability.

Digital platforms now evolve continuously, while traditional Computer System Validation practices were designed around comparatively bounded releases and stable infrastructure. Cloud-native architectures, managed services, automation, AI, and continuous delivery mean that the system under assurance can change throughout its operating life. Assurance must therefore evolve with the platform so that evidence and oversight remain aligned with its current state.

This evolution does not mean less validation or reduced Quality oversight. It means moving from assurance that depends primarily on retrospective evidence assembly towards an approach in which governed controls and contemporaneous evidence support validation throughout design, deployment, operation, and controlled change. Intended-use assessment, risk-based judgement, exceptions, change impact, and Quality decisions remain essential.

When assurance remains concentrated at the end of delivery, it can lag behind the system it is meant to control. The result is validation backlog, duplicated evidence, brittle release governance, and uncertainty over whether the documented validated state still reflects the operating state. Shift-Left CSV addresses that gap by bringing assurance closer to where systems are designed, changed, and operated.

For regulated organisations, the intended benefits are more current evidence, less retrospective reconstruction, stronger governance, and improved delivery agility where controls are well designed and operated. These are intended outcomes, not measured or assured results of adopting Shift-Left CSV.

The central message is: Qualify for reuse. Inherit where justified. Prove continuously. Qualify shared platform controls before they are offered for reuse; inherit them only after workload-specific intended-use assessment, residual-risk review, change-impact assessment, and exception handling; and continuously generate current evidence for governed review. Quality remains accountable for the assurance decision.

The sections that follow explain how this is achieved; this summary is about why it matters.

The remedy: validation by design

The remedy is not to loosen validation expectations, but to design platform controls and evidence services that support product quality, data integrity, traceability, and change control. Microsoft capabilities can provide implementation patterns for policy enforcement, deployment records, monitoring, identity, evidence storage, and lineage, but each use remains subject to intended-use assessment, configuration and integration evidence, risk review, exceptions, and Quality approval. ISPE GAMP® 5, A Risk-Based Approach to Compliant GxP Computerized Systems (Second Edition) (July 2022), is non-binding industry guidance that supports risk-based practice and critical thinking. FDA's Computer Software Assurance for Production and Quality Management System Software (February 2026) provides recommendations for systems used in medical-device production or

quality management systems under 21 CFR Part 820. Neither source makes this model, its architecture, or any product blanket evidence of compliance.

What changes

Shift-Left CSV changes where validation evidence is created and maintained. Instead of relying only on reconstruction after implementation, appropriately governed platforms can generate evidence during design, deployment, and operation. Three changes define the proposed direction:

Evidence becomes continuous: validation evidence is generated automatically through platform controls, operational telemetry, and governed processes rather than manual documentation activities.

Data lineage becomes systematic: lineage, traceability, and data consistency are captured by the platform through metadata, governed data movement, and controlled interfaces instead of being reconstructed from screenshots, logs, or interviews.

Controls become engineered: policy-driven controls, declarative infrastructure, versioned interfaces, automated testing, and observable integrations can support compliance and reduce reliance on manual validation activities where risk and intended use justify it.

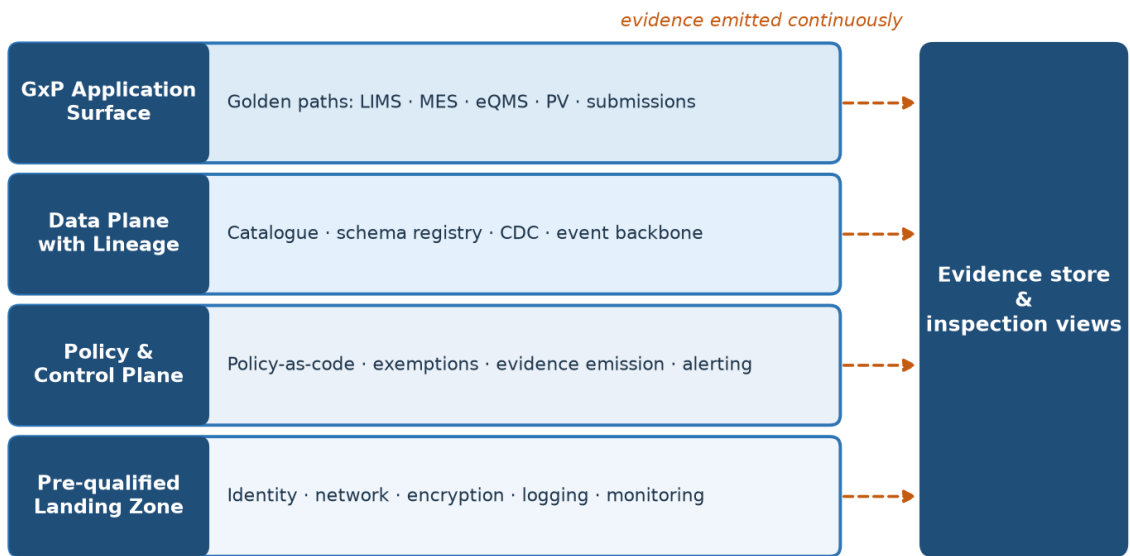
As a result, evidence capture can become a governed platform capability rather than an artefact reconstructed only for periodic review.

What it means for each leader

For executive sponsors, Shift-Left CSV is an enterprise operating-model choice that can reduce delivery friction only if assurance remains effective. Business Quality, GxP, and Operational Excellence leaders define the control intent, process fit, residual-risk decisions, and change disciplines. IT Quality, GxP, and Data Platform leaders implement and operate reusable patterns, including Microsoft technologies where appropriate, while preserving workload accountability and inspection-ready evidence. Successful adoption therefore depends on shared ownership, explicit governance, and Quality decisions throughout the system life-cycle.

Call to action

- **Align on the target assurance model:** jointly define which validation evidence should be generated by platform controls, which artefacts remain document-based, and how both will be presented to auditors.
- **Prove the outcome on a representative platform:** validate the Shift-Left CSV approach through one representative GxP workload, demonstrating that compliance, traceability, and inspection-ready evidence can be generated continuously by design, while reducing manual validation effort and improving delivery agility.
- **Institutionalise the operating model:** establish shared ownership across technology, CSV, and data platform leadership so that policies, evidence stores, lineage services, and platform golden paths are governed as controlled enterprise capabilities.

Figure 8.1 — Shift-Left Reference Architecture (single-page view)

Controls and evidence are built into every layer — validation scales with delivery.

Figure 1 — The shift-left model at a glance

2. Context — Why Traditional CSV Needs to Evolve

The assumptions behind traditional CSV — monolithic systems, on-premises infrastructure, and infrequent change — no longer fully align with continuously evolving cloud platforms; CSV must progressively evolve towards risk-based assurance that keeps evidence and oversight aligned with the operating state.

Traditional Computer System Validation was developed around assumptions that made sense for earlier generations of GxP systems:

- applications were comparatively monolithic
- infrastructure was largely on-premises
- releases were infrequent
- validated state could be captured through a bounded set of specifications, test scripts, installation qualification, operational qualification, performance qualification, and controlled change records.

GAMP 5 First Edition established a risk-based lifecycle model for computerized systems, but in practice many organizations operationalized it through sequential stage gates and document-heavy validation packages.

Why it breaks in the cloud

That model becomes increasingly strained when the system under control is no longer a fixed application stack, but a cloud-native platform composed of managed services, infrastructure-as-code, APIs, data pipelines, containerized workloads, and continuously deployed components.

Three failure modes

The first failure mode is validation backlog.

When every platform change, service upgrade, configuration adjustment, or pipeline modification is treated as a discrete validation event, the CSV function cannot keep pace with engineering velocity. The backlog is not simply administrative; it becomes a control risk because teams either delay beneficial changes or proceed with evidence that trails the actual state of the platform.

The second failure mode is frozen-stack syndrome:

Regulated teams preserve an ageing “validated” environment because requalification is perceived as too costly or too slow. That may appear conservative, but it often increases operational risk by discouraging patching, modernization, resilience improvements, and more secure managed-service patterns.

The third failure mode is evidence drift:

In a cloud platform, the qualified state and the runtime state are constantly moving. Autoscaling, ephemeral compute, managed-service updates, policy changes, identity changes, and data-pipeline

evolution can all alter the operational reality without producing a traditional validation packet. IQ/OQ/PQ artefacts remain useful for specific bounded systems, but they are increasingly inadequate as the primary evidence model for elastic infrastructure. An installation qualification screenshot or static configuration export may be obsolete within hours if the underlying resources are redeployed from templates, rebuilt by pipelines, or modified by provider-managed updates. The artefact may prove what existed at a point in time; it does not prove that the intended controls remain continuously effective.

The economic cost

The economic consequences are material. Late-stage validation compresses release windows, extends time-to-market, and turns compliance into a queue rather than an enabling capability.

Engineering teams spend effort producing retrospective evidence instead of improving the platform. CSV teams spend effort reconciling documents to systems rather than assessing whether controls are designed well and operating effectively.

Cloud economics also suffer: organizations may keep duplicate “validated” stacks running for fear that rebuilds or upgrades will trigger revalidation, creating avoidable spend on idle or underused environments]. The opportunity cost is equally significant: teams defer automation, analytics, AI-enabled data products, and modern integration patterns because the validation burden appears disproportionate to the change.

The regulatory signal

The regulatory and industry signal is that the model is moving. ISPE GAMP 5 Second Edition, published in July 2022, keeps the risk-based framework but modernizes its application for service providers, cloud computing, iterative development, automation, and critical thinking. It explicitly emphasizes the role of knowledgeable subject matter experts in determining fit-for-purpose assurance approaches, rather than defaulting to linear, exhaustive documentation.

ISPE guidance on Computer Software Assurance and critical thinking reinforces the same direction: assurance should focus on confidence that software is fit for intended use, with effort proportional to risk. FDA’s Computer Software Assurance (CSA) guidance — finalised in 2025,” similarly describes CSA as a risk-based, least-burdensome approach for establishing and maintaining confidence that software performs as intended. EU Annex 11 remains anchored in requirements such as validation, data integrity, audit trails, access control, and change control; the 2025 draft revision of EU Annex 11 — and the new draft Annex 22 on Artificial Intelligence — extend these principles to contemporary digital and AI operating models. [expected finalisation ~mid-2026]

The implication is clear: CSV must evolve from a late-stage document gate into an architectural assurance model.

The shift-left thesis begins there: move validation evidence upstream, embed it into the platform, and make compliance an engineered property of how GxP cloud systems are designed, deployed, and operated.

3. Defining Shift-Left CSV

Shift-Left CSV relocates validation evidence to design, deployment, and runtime — preserving regulatory rigour while making it executable in cloud-native environments.

Shift-Left Computer System Validation is an assurance model in which validation evidence is generated during design, deployment, and operation rather than assembled primarily through retrospective document review. It brings control and evidence generation closer to the architecture, pipelines, policies, and services that create and operate the GxP system. Continuously operating controls and their evidence support the validated state; they do not establish it on their own. The validated state depends on that evidence together with intended-use assessment, risk-based judgement, controlled change, exception management, and Quality oversight throughout the system life-cycle. The platform, policies, and evidence store that support this assurance are themselves governed, access-controlled, and validated as appropriate to their intended use.

What it is. Shift-Left CSV establishes engineered controls as the primary mechanism for demonstrating compliance: rather than relying predominantly on procedural activities, technical controls continuously enforce regulatory expectations while simultaneously producing objective validation evidence. Shift-Left CSV treats validation as a property of the architecture. Control objectives are translated into technical patterns, platform standards, policy definitions, pipeline checks, access rules, logging requirements, and evidence records. Controls are expressed as code wherever possible, so they can be versioned, reviewed, tested, approved, deployed, and monitored like any other controlled artefact. Evidence is emitted continuously from normal platform activity: infrastructure deployments, policy evaluations, access events, configuration changes, data movements, test executions, release approvals, and operational health signals. This creates a living evidence model that is closer to the real system state than a static validation package produced after implementation.

What it is not. Shift-Left CSV is not a rebrand of Computer Software Assurance. CSA provides an important risk-based philosophy for establishing confidence that software is fit for intended use, especially by applying critical thinking and proportional effort. Shift-Left CSV is the architectural and operating model that makes that philosophy executable in cloud-native environments. It is also not a justification for skipping validation, reducing QA authority, or replacing human judgment with automation. Automated controls can produce stronger evidence, but the control intent, risk classification, acceptance criteria, and release decision remain governed responsibilities. Finally, it is not a purely technical initiative owned by engineering. It requires joint ownership across CSV, QA, platform engineering, application teams, security, data governance, and business process owners.

Relationship to adjacent concepts. Shift-Left CSV operationalises CSA by providing the platform mechanisms that allow risk-based assurance to be generated continuously rather than reconstructed late. It aligns with GAMP 5 critical thinking by asking which controls matter, where they should live, and what evidence is sufficient for the intended use and patient-risk profile. It borrows from DevSecOps the principle that quality, security, and compliance controls should be embedded in pipelines rather than inspected only at release. It borrows from platform engineering the idea of golden paths: approved, reusable patterns that make the compliant path the easiest path for

delivery teams. It borrows from SRE the discipline of observability, service-level objectives, incident learning, and operational evidence. But Shift-Left CSV is distinct because its design centre is regulated assurance for GxP systems, not only delivery speed, resilience, or engineering efficiency.

Five principles of Shift-Left CSV.

1. **Trusted Foundation (Pre-qualified platform):** GxP workloads inherit baseline qualification from a continuously governed landing zone that provides approved identity, network, encryption, logging, resilience, and monitoring capabilities.
2. **Compliance by Design (Policy-as-code controls):** regulatory expectations, SOP requirements, and architectural standards are translated into machine-evaluable policies that can prevent, detect, or remediate non-compliant configurations. These controls are governed-by-design: they operate within an approved governance framework that defines ownership, risk acceptance, segregation of duties, and Quality oversight.
3. **Controlled Change (Immutable and declarative infrastructure):** environments are built from version-controlled definitions, not manual configuration. Every change is reviewable, reproducible, and traceable to an approved source.
4. **Continuous Assurance (Continuous evidence emission):** deployments, control evaluations, access changes, test results, data movements, exceptions, and operational events generate evidence automatically into a governed, tamper-evident evidence store. This governed evidence repository provides immutability, version history, provenance, and complete traceability from regulatory requirement through technical control to operational execution.
5. **Complete Traceability (Lineage-by-substrate):** data provenance is captured by the data plane itself through catalogues, metadata, schema registries, event logs, and controlled pipelines, rather than reconstructed after the fact during inspection.

Together, these principles establish the foundation for an engineering-led validation operating model. Paper 2 (Technical Reference Architecture) describes the platform architecture and technical capabilities that realise these principles; the operating model and governance that sit above them are set out in the next section.

4. Operating Model & Governance

Shift-left succeeds only as an operating model: QA/CSV retains ownership of control intent and final assurance, Platform Engineering owns the validated substrate, Application Engineering owns workload implementation, and the Business Process / System Owner remains accountable for intended use and business-process fit.

Shift-Left CSV succeeds only when it is treated as an operating model, not as an engineering automation project. The model depends on four roles with distinct but interlocking responsibilities. QA/CSV retains ownership of control intent, risk rationale, validation strategy, exceptions, and final assurance attestation. Platform Engineering is the custodian of the validated substrate, while Application Engineering implements and operates workloads through approved golden paths and remains responsible for application-level controls, testing, and evidence. The Business Process / System Owner remains accountable for intended use, user requirements, business-process fit, operational ownership, and acceptance of the system for its intended business use. Reusable platform controls may be inherited, but they do not transfer these accountabilities or weaken QA/CSV authority.

Activity	QA/CSV	Platform Engineering	Application Engineering	Business Process / System Owner
Policy authoring	A/R for control intent	R for technical implementation	C on workload impact	C on intended use, business-process impact, and user requirements
Policy approval	A	C	C	C on intended-use and business-process fit
Exemption granting	A/R	C/R for technical risk	R for request and mitigation	C on business impact, intended use, and compensating measures
Evidence review	A/R	R for evidence generation	C	C on intended-use and business-process evidence
Periodic review	A/R	R for dashboards and metrics	C	R for confirming continued intended use and business-process fit
Incident response	C/A for quality impact	R for platform restoration	R for application impact	R for business-process impact assessment and operational response
Audit support	A/R for assurance narrative	R for platform evidence	R for workload evidence	R for intended-use, business-process, and operational evidence

Governance forums

The governance model should be lightweight but explicit. A **Control Board**, chaired jointly by QA/CSV and Platform Engineering, should meet monthly to approve new controls, retire obsolete controls, review control performance, and decide when a policy can move from audit to deny. An **Exemption Review** forum should meet weekly during early adoption and bi-weekly once stable; it reviews open exemptions, ageing, compensating controls, and expiry discipline. A **Technical Design Authority** should meet weekly to review new golden paths, architectural deviations, reusable platform patterns, and changes that may affect the validated substrate. A quarterly **Assurance Review** should bring together QA/CSV, platform, application, security, data governance, and business owners to review evidence completeness, incidents, regulatory observations, and roadmap priorities.

The cultural shift

The cultural shift is substantial. In the traditional model, CSV is often perceived as a gate at the end of delivery. In the shift-left model, CSV becomes an enabling function that defines the guardrails early enough for teams to move quickly within them. The review focus moves from “is the document complete?” to “is the control operating, evidenced, and linked to risk?” Platform Engineering must accept that regulated platform capabilities are controlled products, not informal shared services. Application teams must accept the golden path as the default route to speed, not a constraint on autonomy. The goal is not less governance; it is governance embedded where work happens.

KPIs that signal success

KPIs should show whether the model is improving both delivery and assurance. Useful measures include deployment frequency for GxP workloads, lead time for validated changes, exemption count and ageing, percentage of controls with automated evidence, evidence completeness by workload, policy drift rate, failed control evaluations, audit findings closure time, incident recurrence, and the percentage of workloads consuming approved golden paths. A healthy model should show increasing deployment frequency without increasing open deviations, declining exemption ageing, faster audit-response preparation, and higher evidence completeness. If delivery accelerates but exemptions accumulate, the model is becoming a bypass mechanism. If evidence completeness improves but deployment frequency does not, the platform may be over-controlled. The operating model must continuously balance both dimensions: faster compliant delivery and stronger, more current assurance.

5. Adoption Roadmap

A phased 12–18 month path — Establish, Prove, Scale — makes the transition real and risk-managed, and avoids the anti-patterns that derail similar programmes.

Phase	Objectives	Key deliverables	Success criteria	Primary risk to watch
Phase 1: Establish 0–3 months	Select one representative GxP workload and establish the minimum viable validated substrate. Align QA/CSV, Platform Engineering, Application Engineering, security, data governance, and the Business Process / System Owner on the target assurance model. Document the current baseline for assurance, validation effort in person-hours, and change lead time, and agree the criteria that will be used for comparison during the pilot.	Named pilot workload; pre-qualified landing-zone baseline; initial policy catalogue with 5–10 priority controls; evidence model and evidence-store design; initial RACI; draft inspector narrative explaining the shift-left evidence model; documented baseline for assurance, validation effort/person-hours, and change lead time; pre-agreed comparative criteria for the pilot.	Landing zone deployed in a controlled non-production environment; priority policies assigned in audit or deny mode as appropriate; evidence records generated for deployments, access, policy evaluations, and configuration state; pilot scope, documented baseline, and comparative criteria approved by QA/CSV, the Business Process / System Owner, and technology leadership.	Over-scoping the foundation. If Phase 1 tries to solve every control, platform, and workload at once, the programme will lose momentum before it produces evidence.
Phase 2: Prove 3–9 months	Migrate or onboard the first GxP workload onto the validated substrate, run a parallel validation in shadow mode, and test whether shift-left evidence can withstand inspection-style scrutiny.	Pilot workload deployed through approved golden paths; policy-as-code controls linked to requirements and risks; automated test and deployment evidence; exception workflow; mock inspection package; documented comparison of traditional and Shift-Left approaches against the agreed baseline and criteria for assurance, validation effort/person-hours, and change lead time.	Mock inspection completed with documented findings and responses; evidence completeness meets the agreed threshold; critical controls operate without manual reconstruction; results demonstrate at least equivalent assurance with no increase in material Quality risk and document material improvement in validation effort/person-hours and/or change lead time against the baseline; open deviations do not	Shadow validation becoming duplicate validation. If teams simply run traditional CSV plus shift-left evidence indefinitely, the model will increase burden rather than prove a better operating model.

			increase; exemptions are time-bound and reviewed.	
Phase 3: Scale 9–18 months	Extend the validated substrate from pilot to platform capability. Onboard additional workloads, industrialise golden paths, and retire legacy validation artefacts where they are clearly superseded by controlled platform evidence.	Expanded golden-path catalogue; reusable policy initiatives; production evidence dashboards; periodic review automation; workload onboarding playbook; decommissioning plan for superseded artefacts; governance cadence embedded into operating forums.	Multiple GxP workloads onboarded; percentage of controls with automated evidence increases; audit preparation time decreases; exemption ageing decreases; platform adoption grows without material increase in quality incidents or audit findings.	Platform bottleneck. If all changes require central platform-team intervention, the validated substrate becomes another gate rather than an accelerator.

What not to do — transformation anti-patterns

- **Do not boil the ocean:** start with one meaningful GxP workload and a small number of high-value controls, not an enterprise-wide control catalogue.
- **Do not create policy committees without engineering input:** policies that cannot be implemented, tested, monitored, or supported will remain theoretical controls.
- **Do not build evidence stores that nobody reads:** evidence must feed review routines, dashboards, inspections, and release decisions, or it becomes another archive.
- **Do not leave exemptions open-ended:** every exemption needs an owner, risk rationale, compensating control, expiry date, and review cadence.
- **Do not retire traditional artefacts before the replacement evidence is accepted:** phase out legacy validation documents only when the shift-left evidence is mapped, approved, and inspection-ready.

6. Conclusion and Way Forward

Shift-Left CSV is a governed operating-model change: it brings assurance closer to design and operation without transferring Quality accountability to architecture, automation, or a vendor product.

Book 1 has set out the case for moving from retrospective evidence assembly towards reusable, governed controls and more current evidence. Adoption is not a slogan or a one-off technology deployment; it is a phased organisational change in decision rights, engineering practices, evidence review, exception management, change control, and Quality oversight.

- **Book 2 — Technical Reference Architecture: architects and platform teams can translate this operating model into architecture principles, policy-as-code, lineage and consistency patterns, and a reference architecture, including Microsoft implementation patterns that must be assessed in context.**
- **Book 3 — Regulatory & Implementation Guide: CSV, Quality, and Regulatory teams can develop the regulatory mapping, implementation and inspection-readiness approach, treatment of AI workloads, and detailed risks and mitigations.**
- **Practical next step: convene the five audience groups around one representative GxP context to agree intended-use boundaries, decision rights, reusable platform controls, workload-specific obligations, and the evidence needed before implementation proceeds.**

The destination is not compliance by architecture, but more current and repeatable risk-based assurance. Engineered controls provide evidence; people remain accountable for intended use, residual risk, exceptions, change impact, and the final Quality decision. Books 2 and 3 carry that vision into technical and regulatory implementation.